Particle NeRD

Learn particle motion.
Reconstruct body reactions analytically.

Replacing the expensive MPM update while retaining interaction with rigid ground and moving bodies.

Latest experiment progress →

REFERENCE DATA · LEARNED DYNAMICS

LOCAL CODE

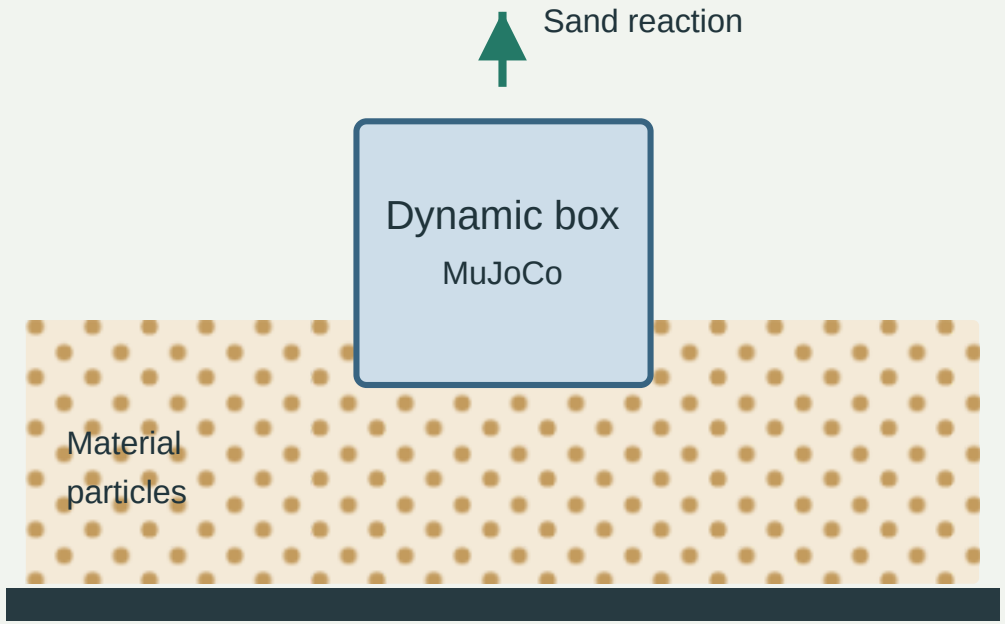
A sand bed, rigid ground, and one dynamic body

MuJoCo advances the rigid body.
MPM advances the sand and supplies contact impulses.

Keep Rigid-body dynamics, rigid-body contacts, and Newton’s coupling interface.

Replace The expensive material update with learned motion and analytical reaction reconstruction.

Dynamic collection is being implemented with MJWarp–MPM coupling. Earlier results use a prescribed box; see the progress page.



Fixed rigid ground beneath the sand

Particle motion and body reaction must remain consistent.

Source: Newton coupled MPM–MuJoCo example. Illustration is schematic.

LOCAL CODE

Common geometry; different motion constraints

Newton converts supported shapes into collider meshes and rasterizes contact information onto the MPM grid.

COLLIDER	GEOMETRY AND CONTACT PROPERTIES	MOTION	REACTION DESTINATION
Rigid ground	Mesh, distance, normal, friction	Fixed	Fixed environment
Prescribed tool	Same representation	Externally supplied	Measured reaction; pose remains prescribed
Dynamic body	Same representation	Rigid-body solver	Rigid-body solver

Implication for NeRD Use shared collider features. Preserve each collider’s identity, motion, and material properties; separate “ground” and “shovel” networks are not required.

Sources: shape conversion · collider queries · rasterization and coupling

SLIDE DECK

The cable model learns motion corrections

01

Collect

Record cable trajectories from VBD under randomized manipulation and contact.

Segment position, orientation, and motion history.

02

Train

Start from a recorded state. Predict corrections to motion under inertia and gravity.

Supervise one-step corrections using the next VBD frame.

03

Infer

Apply predicted corrections, enforce constraints, and advance from the model's own state.

VBD is absent from the cable inference loop.

```
baseline motion + learned correction → constraint enforcement → next state
```

Source: cable representation and training. The deck does not specify the exact baseline integration formula.

CABLE SLIDES 23–24

The cable reconstructs forces after predicting motion

Motion history
Three segment poses → accelerations

→

Known physics
Mass, inertia, gravity and rod bending law

→

Chain solve
Joint and contact force / torque balance

→

Body reaction
Sum opposite contact forces and torques

What makes it tractable

Fixed chain connectivity creates a block-tridiagonal system. Geometry restricts where external contact forces may act.

What the network supplies

Motion only. The reconstruction uses physical equations and softly enforced rod moments; no force network is required.

Transferable principle

Learn the expensive motion update; reconstruct reactions with known mechanics.

Source: force reconstruction · chain implementation. The cable implementation applies delayed, capped body reactions.

METHOD DISTINCTION

Constraint solving and force reconstruction are related

XPBD

Solve for constrained motion

Correct predicted positions using physical constraints. Constraint multipliers provide force estimates.

Prediction → constraint solve → motion + forces

CABLE NeRD

Reconstruct forces from motion

Use predicted motion, contact geometry, and material laws to solve for consistent reactions.

Learned motion → balance equations → forces

A collision push-out is insufficient

If NeRD already predicts a supported, nonpenetrating state, the final push-out can be zero while the physical support force is nonzero.

Sources: XPBD, Macklin et al. (2016) · cable slide 23

PAPER + RELEASED CODE

L-GBND separates material, tool, and static nodes

NODE TYPE IN RELEASED CODE	POSITION UPDATE	ROLE
Terrain / type 0	Current position + learned displacement	Deforming material
Tool / type 1	Current position + prescribed action	Moving contact geometry
Wall / type 2	Unchanged	Static environment

Useful precedent. A graph can represent material–tool interaction with distinct node roles.

Missing for our task. The released position predictor does not return a reaction wrench to a freely moving body.

The paper’s region-boundary normals represent omitted stationary terrain. They do not specify an explicit rigid-ground collision solve.

Sources: L-GBND, Sections III-A–D · released node updates

DESIGN

One material model, explicit collider information

PARTICLE STATE

Motion and neighborhood

Positions, velocities, masses, and relative positions and velocities of nearby material samples.

Record APIC velocity gradients; test whether they are needed as model state.

CONTACT INPUTS

Geometry and motion

Distance, surface normal, relative surface velocity, friction, and collider identity.

Include both ground and body features where both are relevant.

MODEL OUTPUT

Particle motion

A learned displacement correction to an explicit gravity-and-velocity baseline.

No learned body-reaction output. No MPM solve at inference.

Update every material particle

Use a small complete sand bed with local graph connections. No region selector; keep material, timestep, and resolution fixed initially.

Proposal informed by cable NeRD and Newton's APIC state and collider inputs.

OPEN

What replaces the cable’s chain equations?

```
mass × particle acceleration
= gravity force + internal material force + collider contact forces
```

Known from motion and geometry

Acceleration, mass, gravity, potential contacts, contact normals, and collider motion.

Unknown

Internal stress transmission and the separate contact reactions on ground and each body.

Candidate reconstruction

Solve local balance on a coarse grid or particle neighborhood graph, constrained by material response and admissible contact forces.

Compression, friction, and material assumptions must be defined. Neighborhood edges are not cable joints.

An initial particle-graph reconstruction has now been tested; see slides 16–18. A suitable granular material model remains an open question.

BEFORE NETWORK TRAINING

Test reconstruction using MPM’s own motion

MPM trajectories

Motion history + geometry
+ known material parameters

→

Analytical reconstruction

Estimate body reaction
without MPM force labels

→

Reference comparison

Compare with MPM impulses
and body torques

EXPERIMENT	WHAT IT ESTABLISHES
Sand at rest on rigid ground	Recovered ground support balances sand weight.
Box supported by sand, clear of rigid ground	Recovered sand force supports the box’s weight.
Box pressing into and sliding through sand	Reconstruction captures changing normal load, friction, and torque.

Use an MPM iteration cap of 25–50. Check static force balance and stable motion; increasing the iteration cap is not a prerequisite.

PLANNED DATASET

Record motion, contacts, and validation references

Generate varied MPM episodes

- Settling and static support.
- Vertical pressing and release.
- Lateral pushing and sliding.
- Different starting depths and body orientations.

Use prescribed trajectories for motion-model experiments and dynamic-body episodes for coupling evaluation.

Save each timestep

- Particle position, velocity, mass, and APIC gradient.
- Body pose, velocity, and contact material parameters.
- Material configuration, timestep, and resolution.
- MPM body impulses and torques for validation.

Split by episode

Reserve complete motions and initial conditions for validation and testing. Adjacent frames from one episode must not cross dataset splits.

MPM baseline: 25 iterations maximum, tolerance 1e-4, following the G1 configuration. Reaction labels are for validation only.

IMPLEMENTATION

Train on MPM transitions; select by rollout accuracy

Recorded state at t
Particle state
+ collider inputs

→

Baseline motion
Current velocity
+ gravity

→

NeRD correction
Particle interactions
+ contact response

→

MPM state at t + 1
Supervised
motion target

```
target displacement correction = MPM next position − baseline next position
```

Initial training recipe. One-step supervision with independent position and velocity perturbations; test noise scale and loss choice.

Checkpoint selection. Repeated model rollouts with identical collider trajectories. Measure drift, penetration, and material redistribution.

Proposed modules: `particle_model.py` · `train_motion.py`. The cable recipe motivates these choices; particle stability remains to be tested.

IMPLEMENTATION

Close the loop with an analytical reaction step

MuJoCo rigid-body update

Pose and velocity under applied body wrenches

→

Particle NeRD update

Baseline + learned correction + geometric constraints

Newton coupling interface

Per-body force and torque for the next coupled step

←

Analytical reconstruction

Particle motion + material model + contact geometry

↑ Apply reconstructed reaction

↓ Reconstruct after the particle update

Retain collider identity throughout

Sum each contact reaction onto the correct body. Compute torque about that body's center of mass; keep fixed-ground reactions separate.

Proposed adapter: `solver_particle_nerd.py`. Integration target: Newton's proxy-wrench interface.

RESULTS ON SLIDES 15–18

Implementation milestones

MILESTONE	TASK	SUCCESS CRITERIA
1 · Data collection	Record particle motion and body reactions using an MPM iteration cap of 25–50.	Particle and body trajectories remain stable. Support forces balance weight in static tests.
2 · Force reconstruction	First check whether body reactions are uniquely determined. Then reconstruct forces using explicit material and contact relationships.	Estimated reactions agree with MPM and take less time to compute than the MPM update.
3 · Motion prediction	Train a graph neural network to predict particle motion from MPM trajectories.	Predicted trajectories match MPM on test episodes over multiple timesteps.
4 · Coupled simulation	Use NeRD and reconstructed reactions in Newton to simulate particles and moving bodies.	Body motion and sand deformation remain close to MPM without running the MPM solver.
5 · Larger scenes	Increase the number of particles while updating every particle at each timestep.	The complete simulation runs faster than MPM while maintaining the required accuracy.

Define acceptable motion and force errors before evaluation. Measure runtime for the complete simulation, including contact queries and force reconstruction.

COLLECTION CHECKS PASSED

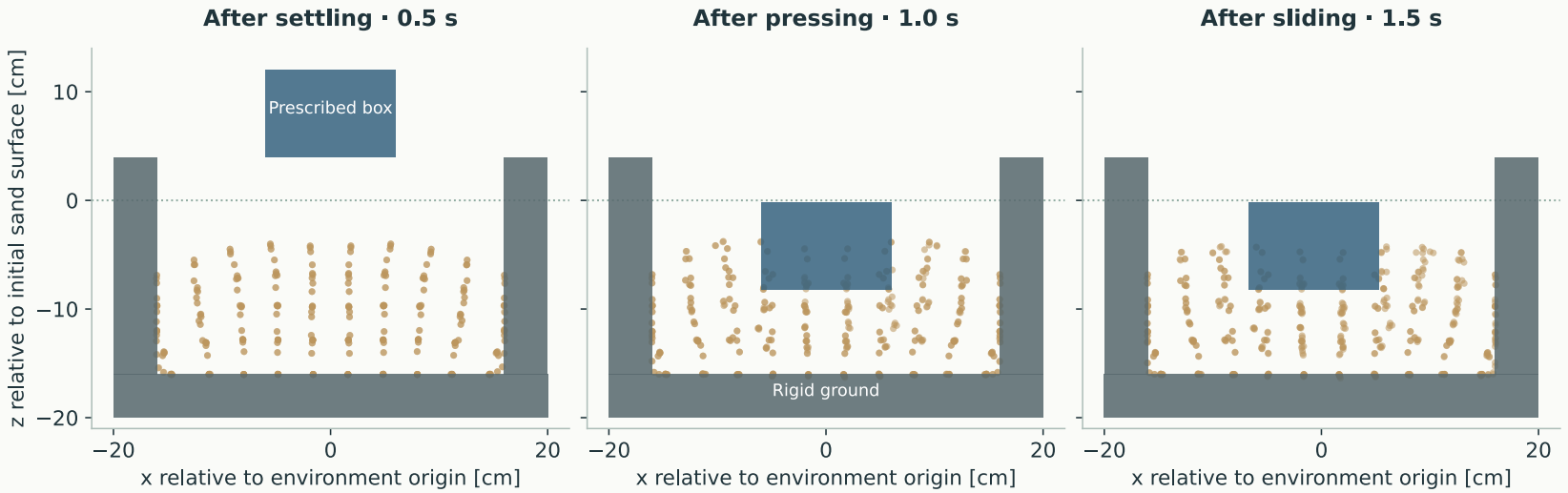
Reference data collected from parallel MPM simulations

2
parallel environments

500
particles per environment

4 × 2 s
recorded episodes

1,600
recorded transitions



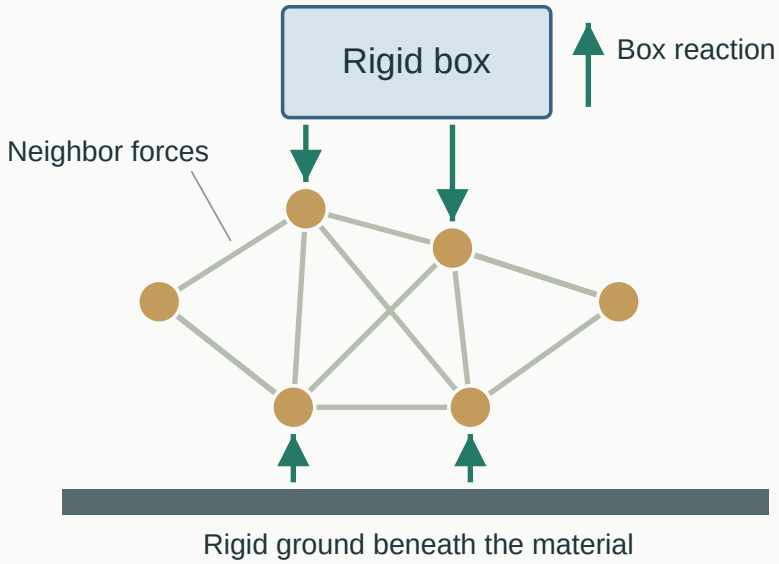
Actual recorded positions, projected onto the x–z plane. Front and rear walls are omitted from the illustration.

Verified: recording alignment, episode resets, particle retention, settling, and total linear momentum balance.

25 MPM iterations maximum · tolerance 1e-4 · timestep 5 ms · momentum residual 0.00084% of the reference scale.
Small validation dataset; no model training. Recorded data retained locally · Implementation and run commands

EXPERIMENTAL ASSUMPTION

Estimate body reactions from recorded particle motion



Schematic of the reconstruction assumption; not an MPM force network.

1. Measure acceleration
Use the change in each particle's velocity.

2. Solve particle force balance
Allow compressive forces between neighbors and friction-limited contact forces.

3. Sum reactions on each body
Apply equal-and-opposite contact forces and their moments about the body center.

Key assumption

A small force penalty selects one solution when motion admits several force distributions. This does not reproduce Newton's granular material law.

Inputs: MPM particle motion, mass, gravity, and rigid-box geometry. MPM reaction forces are withheld until evaluation.
No neural model is used. All particles participate; ground and moving boxes use the same contact representation.

SUCCESS CRITERIA NOT MET

The current reconstruction misses the box reaction



One recorded episode. Positive force points upward.
Reconstruction markers show only the four evaluated times.

38.5%

relative body-force error
across 16 sampled frames

79.9 N

moving-box force RMSE
across those 16 frames

7.0 s

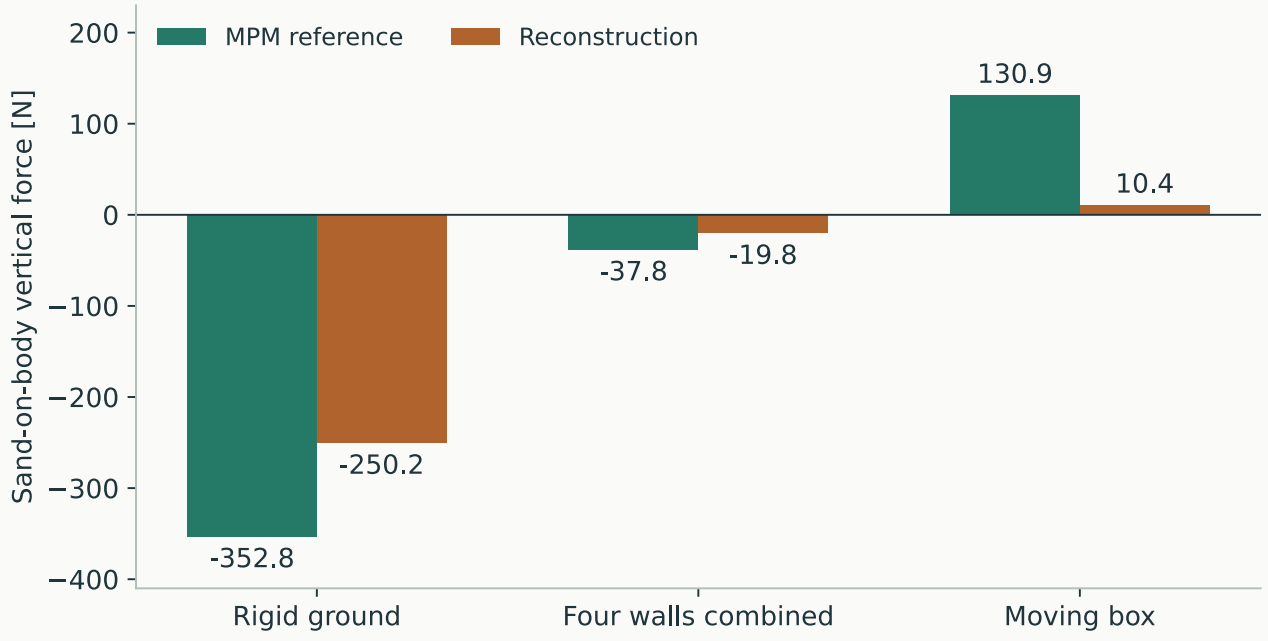
median CPU reconstruction time
per sampled frame

16 / 16 solves reached the iteration limit.
These are provisional estimates.

Relative error = norm of all estimated-minus-reference body forces / norm of all reference body forces.
Four frames per episode, six bodies per frame; force error includes all three components. Full evaluation results (local file).

MECHANICS STILL UNRESOLVED

Correct total force can hide incorrect body forces



Episode 0 at 1.0 s: force exerted by sand on each body.
Positive is upward; negative is downward.

Total vertical force on all bodies

−259.65 N MPM

−259.65 N reconstruction

The totals agree, but the estimated upward force on the moving box is much too small.

Implication: satisfying particle force balance does not validate the force on each body.

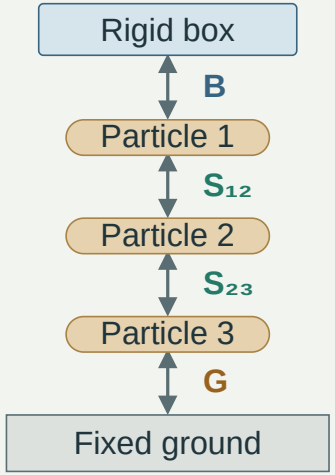
Next: check whether the full particle equations uniquely determine body reactions, before choosing a force-selection rule.

Optimization error and the material-force assumption are both unresolved. A separate one-frame diagnostic converged under adjusted tolerances but retained 44.8% body-force error; it is not a full reevaluation.

ONE-DIMENSIONAL EXAMPLE

Equations link the rigid body, particles, and ground

Upward is positive. Each contact force appears with opposite signs on the two objects it connects.



B, S_{12}, S_{23}, G: compressive force magnitudes at successive contacts.

Rigid box	$m_b\, a_b = U + B - W_b$
Particle 1	$m_1\, a_1 = S_{12} - B - W_1$
Particle 2	$m_2\, a_2 = S_{23} - S_{12} - W_2$
Particle 3	$m_3\, a_3 = G - S_{23} - W_3$
Fixed ground	$0 = R - G - W_g$

m = mass; a = vertical acceleration; W = weight (m g).
U: external actuator force on the box. **R:** external support force on the ground. Both are signed upward.

Add all five equations: $m_b\, a_b + m_1\, a_1 + m_2\, a_2 + m_3\, a_3 = U + R - \Sigma_i\, m_i\, g$

Contact forces cancel in the sum. **The actuator force U and support force R remain.**
With known U and measured box acceleration, B follows from the box equation. With prescribed box motion and unknown U, that equation introduces another unknown.

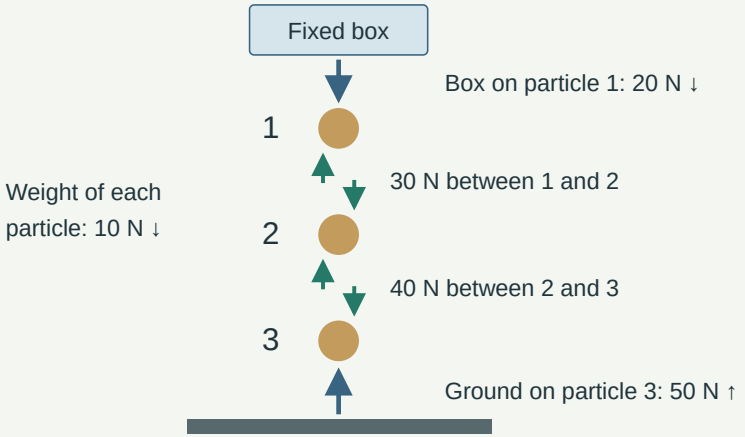
Sum over the box, particles 1–3, and supported ground slab. Vertical translation only; zero ground acceleration does not imply zero contact force.

EXACT FORCE-BALANCE EXAMPLE

Different contact forces can produce identical motion

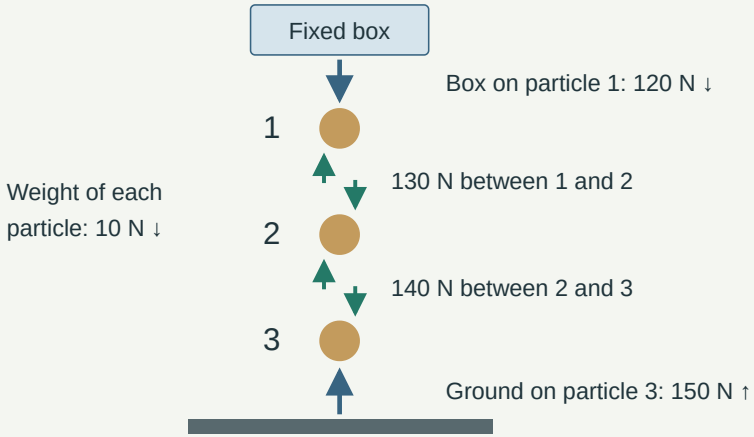
Three stationary particles. Only particle 1 touches the box; only particle 3 touches the ground.

A · Lower compression



Particle 1	30 − 20 − 10 = 0 N
Particle 2	40 − 30 − 10 = 0 N
Particle 3	50 − 40 − 10 = 0 N

B · Add 100 N along the entire chain



Particle 1	130 − 120 − 10 = 0 N
Particle 2	140 − 130 − 10 = 0 N
Particle 3	150 − 140 − 10 = 0 N

Every particle equation is satisfied in both cases. Adding the equations gives ground force − box force = 30 N in either case.

Upward force is positive. Equal-and-opposite internal forces cancel. Identical geometry and zero acceleration do not select A or B.

This example uses force balance and compressive contact only; a material law or a known boundary load could provide the missing information.

NOT YET COMPLETED

Check whether the equations determine each body reaction

1. Assemble all balances

Balance each particle's forces. Internal forces have opposite signs on neighbors. Include contact forces only where contact occurs.

→

2. Look for force changes

Can forces change while every balance remains satisfied and contact constraints remain valid?

→

3. Check body reactions

If those changes alter a body's net force or torque, motion alone has not determined that reaction.

Some ambiguity is acceptable

Internal forces may be nonunique while every body reaction is unchanged. We only need the body reactions to be determined.

Otherwise, identify missing information

Examples: a material force–deformation relationship, material history, or a known applied boundary load. Test whether it resolves the ambiguity.

Correction to the work sequence

This analysis should have preceded the minimum-force implementation. That implementation is an exploratory baseline, not a validated reconstruction method.

Inference requirement: NeRD particle motion + nonlearned reaction reconstruction. No MPM solve at inference.
MPM is used only to generate reference data and evaluate results.

IMPLEMENTED BASELINE

The current estimate omits most material properties

ASSUMPTION	WHAT THE IMPLEMENTATION USES
Particle–particle forces	Particles within 6.5 cm transmit equal-and-opposite compressive forces along their connecting line. No separate tangential force.
Particle–body forces	Particles within 4 cm of a body surface are contact candidates. Forces obey a four-sided approximation of the friction cone.
Choice among possible solutions	A quadratic penalty favors small force coefficients. It supplies no force–deformation relationship.

Current objective minimize: force-balance error² + 0.0001 × Σ force coefficient²

Parameters used directly

Particle mass and particle–body friction. Density affects the recorded particle mass.

Material properties not used directly

Young’s modulus, Poisson’s ratio, internal material friction, plasticity, and material history.

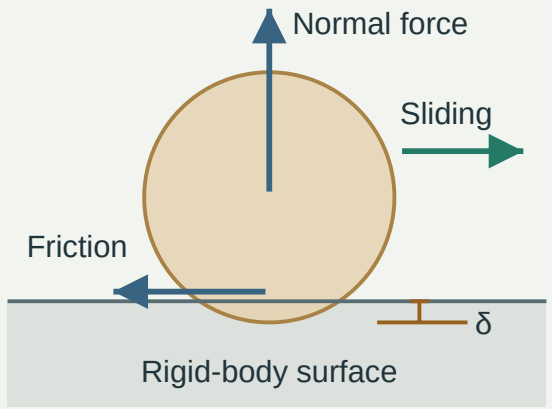
With identical motion, masses, geometry, and body friction, changing an omitted material parameter does not change the estimate. It can affect the result indirectly by changing the motion.

Source: current reconstruction implementation. Small force-balance error does not establish accurate forces on individual bodies.

SOURCE OF ADDITIONAL EQUATIONS

Relate each contact force to deformation and motion

A simple noncohesive DEM contact law supplies force values from overlap, relative velocity, and contact history.



δ = particle–surface overlap

Illustration of a DEM grain.
MPM material points are not physical grains.

Normal force magnitude · active contact

$$F_n = \max(0,\ k_n\,\delta + c_n\,v_{close})$$

Elastic compression + damping; closing speed is positive.

Tangential force vector · before sliding

$$F_{t,trial} = -k_t\,\xi - c_t\,v_t$$

ξ accumulates relative tangential displacement during contact.

Sliding limit

$$\|F_t\| \le \mu\,F_n$$

Cap the trial force and update history when sliding occurs.

Additional equations: force is related to contact deformation and history, rather than selected only by a small-force preference. Separated particles carry no contact force.

k: contact stiffness; c: damping coefficient; μ: contact friction. Ground and moving bodies use the same law with their own geometry and surface velocity.
Source: LAMMPS granular contact laws and particle–wall contacts. The equations show a linear spring–damper model; Hertz–Mindlin is an alternative.

SOURCE REVIEWED

Warp provides a simple DEM contact-force kernel

Linear spring–damper contact, velocity-based friction, and short-range cohesion. No tangential history.

Normal force on particle i

$$F_n = -s\,n$$

$s = k_n\,c + k_d \min(v_n, 0)$

Tangential force magnitude

$$\|F_t\| = \min(k_f\,\|v_t\|, \mu\,|s|)$$

Direction opposes tangential relative velocity.

For particle pairs: c = center distance − 2 × radius; n points toward particle i; v_n = n · relative velocity.

Cohesion	Positive gaps can produce attraction within a cutoff.
Static friction	Zero tangential speed gives zero friction; no stored tangential displacement.
Geometry	Equal-radius particles and a fixed plane; no arbitrary moving-body contacts.

Proposed reuse: hash-grid neighbors and contact-force evaluation. Add rigid-body surface velocity and reaction accumulation. NeRD supplies motion; do not use the example's particle integration.

Source: NVIDIA Warp · example_dem.py. This is a standalone Warp example, not Newton's implicit MPM solver.

PRELIMINARY EVALUATION COMPLETE

Test a contact law on the recorded particle motion

Evaluated objective minimize: force-balance error² + λ × contact-law error²

The second term favors forces predicted by the contact law. λ sets its weight relative to motion consistency.

1. Define effective contacts

Specify reference spacing or effective radii, contact stiffness, damping, and friction. Do not equate MPM spacing with grain overlap.

→

2. Compare contact laws

Evaluate velocity-only friction and a history-based law. Use particle and body velocities; maintain history through each frame.

→

3. Evaluate body reactions

Compare with MPM reference forces on each body, including densely sampled impact intervals and small position perturbations.

Main modeling question

What effective deformation should determine force? MPM particles sample a continuum; their radius is not automatically a grain radius.

Success criteria

Lower body-force error with small balance residual, stable estimates under position errors, and acceptable reconstruction time.

A penetration-only law gives zero support force at zero overlap and zero relative speed. Reference geometry and contact compliance must therefore be defined and tested.

Inference: NeRD motion → nonlearned contact-force reconstruction → Newton rigid-body update. No MPM solve at inference.

Fix parameters before final evaluation; use separate episodes if calibration is needed. The first evaluation did not improve force accuracy; results follow.

ACCURACY DID NOT IMPROVE

The DEM prior increased force prediction loss

52 matched frames from two evaluation episodes. Parameters selected on two different episodes.
Force MSE averages squared error across frames, six bodies, and three force components.

METHOD	FORCE MSE [N²]	BOX FORCE RMSE [N]	PARTICLE BALANCE RESIDUAL	CONVERGED FRAMES
Original baseline	5,290	214.8	0.04%	0 / 52
Zero prior · new basis	6,877	225.2	14.78%	52 / 52
DEM · velocity friction	6,554	223.9	13.86%	52 / 52
DEM · contact history	6,524	224.0	13.92%	52 / 52

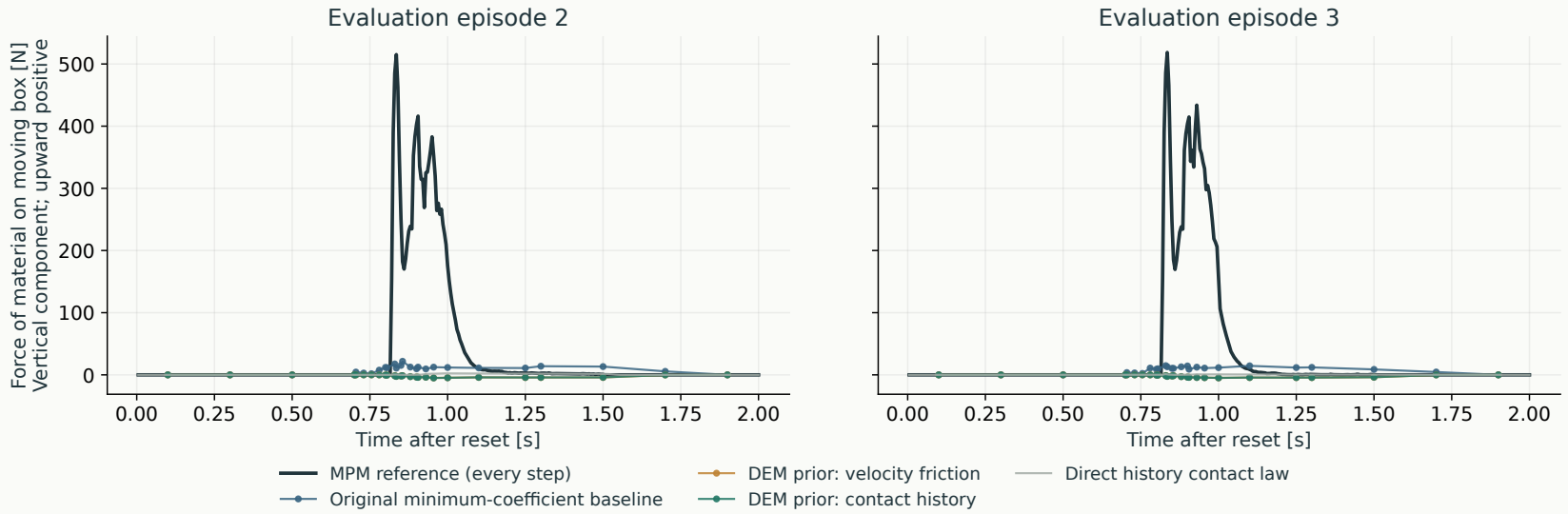
Contact history: 23.3% higher force MSE than the original baseline. The prior improves on the zero-prior variant with the same new force basis, but neither DEM variant meets the particle-balance criterion.

Both selected variants: stiffness 100 N/m; prior weight 0.1. No calibration candidate met both convergence and mean balance residual ≤ 5%.

Box RMSE uses the force-vector error; balance residual is normalized per frame by the particle inertial load after subtracting gravity.
The original baseline reached its iteration limit on all 52 frames. The new solver met its projected-gradient criterion. These are different convergence tests.
Data: DEM experiment (local file) · matched original baseline (local file) · summary (local file).

CONTACT MODEL REMAINS INSUFFICIENT

The reconstructed box force still misses the impact load



Material force on the moving box. Markers show evaluated frames; connecting lines interpolate between samples.

515–519 N
MPM force at each reference peak

−2.5 N
DEM history estimate at those same frames

0.08 / 0.39 N
Force change RMSE for 0.1 / 1 mm position offsets

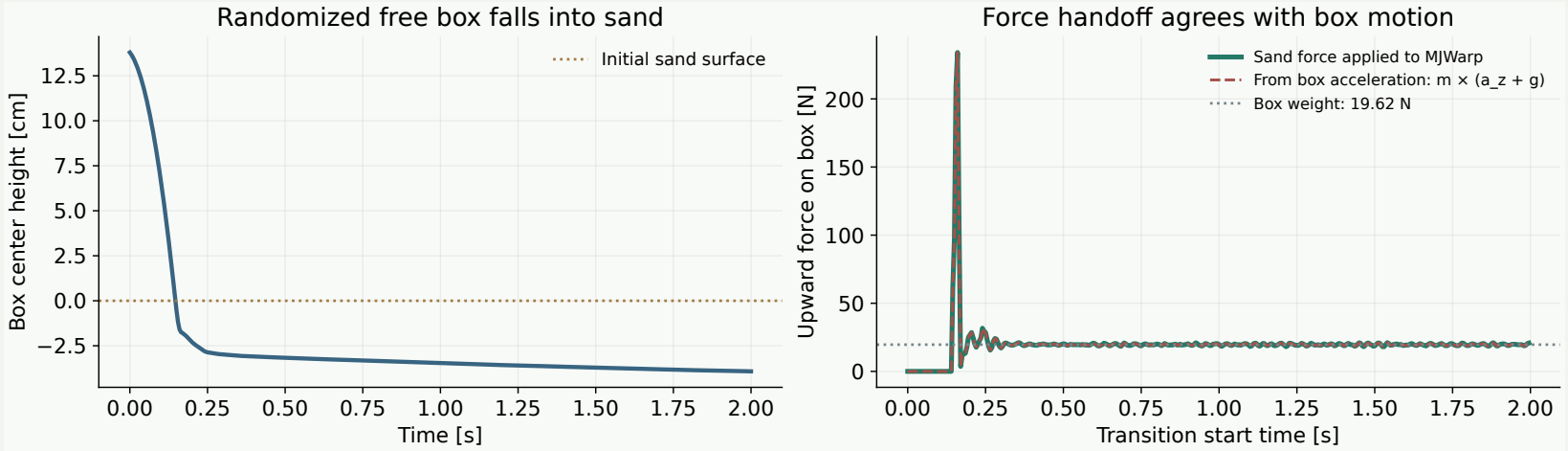
Conclusion: adding this effective contact law did not resolve the missing body loads. Revisit effective contact geometry and material relationships before using the estimate for two-way coupling.

Position sensitivity: 12 matched frames, all body-force components, persistent Gaussian particle offsets per coordinate. Velocities and accelerations unchanged.
Small sensitivity does not establish force accuracy. This experiment uses recorded MPM motion; no MPM solve is retained at inference.

MJWARP + MPM

The box now moves under gravity and contact

Randomize the 2 kg box's initial position, orientation, and velocity. Apply no commands after reset.
MJWarp advances the box; MPM advances the sand; the coupler exchanges motion and reactions.

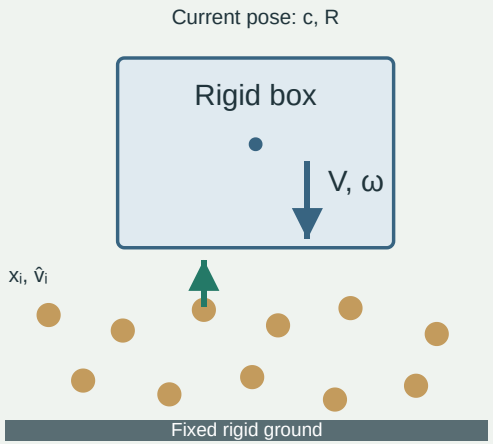


Four episodes, 1,600 transitions: mean box force-balance residual 0.000002 N.
This validates reference coupling. Reconstructed-force accuracy is evaluated separately.

Box balance: mass × velocity change = gravity impulse + applied sand impulse + direct rigid-contact impulse.
Lagged coupling applies the previous MPM reaction; compare matching intervals. All runs use 25 MPM iterations maximum and a 5 ms timestep.
Latest experiment progress · MPM remains absent from the proposed NeRD inference loop.

IMPLEMENTED APPROXIMATION

From particle motion to force on the rigid box



Blue: rigid-body state. Gold: material particles. Green: material/contact quantities.

Inputs and outputs of one force-estimation step

INPUT · particles — Current positions x_i^n, masses m_i, and predicted next velocities $\hat{v}_i^{n+1}$.

INPUT · rigid box — Current center c^n, orientation R^n, linear velocity V^n, angular velocity ω^n; mass m, inertia I, and box size.

OUTPUT — Material-on-box force F and torque τ; contact-corrected box velocities V^+, ω^+ and per-particle impulse $-j_i$.

Also supplied: Δt, gravity g, friction μ, contact offset δ, effective mass scale κ, penetration coefficient β; previous impulses if history is enabled.

> This routine estimates contact loads. Particle positions and box pose are held fixed inside it. Newton advances the rigid-body pose separately.

1 $V^* = V^n + \Delta t\, g$ **2** $\hat{u}_i = \hat{v}_i^{n+1} + G_i(s_i - x_i^n)$

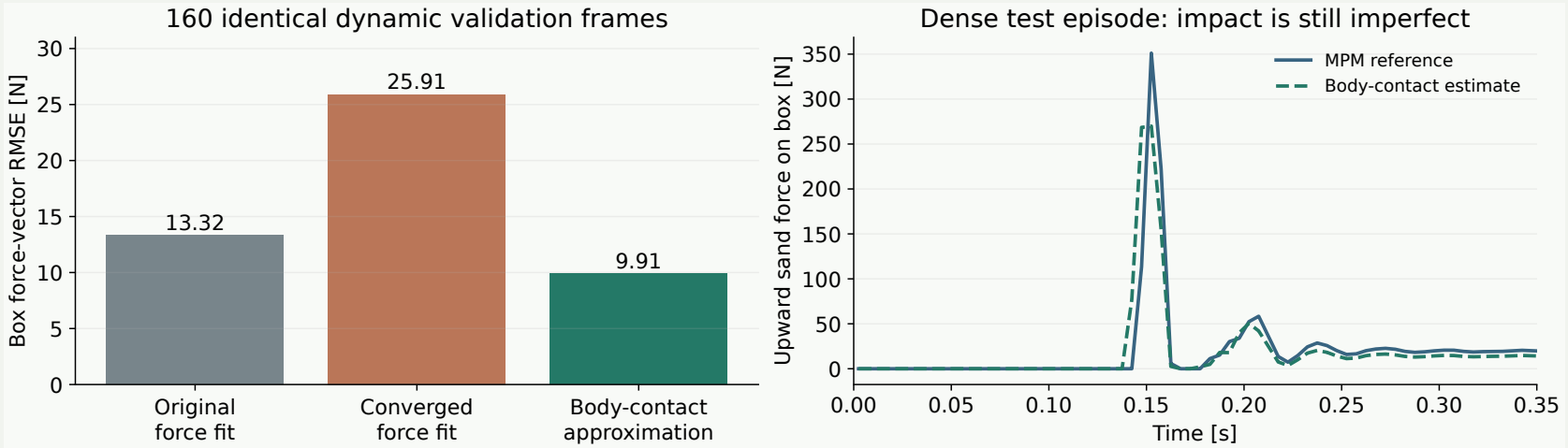
3 Solve unilateral contact + Coulomb friction with effective mass **4** $F = \Sigma j_i / \Delta t$, $\tau = \Sigma r_i \times j_i / \Delta t$

Current experiments use recorded next particle velocities in place of NeRD predictions. Geometry uses current particle positions and box pose.
The estimator runs no MPM solve. It estimates the dynamic-box reaction; fixed ground and wall reactions are not outputs. HTML tabs show the full derivation.

MATCHED-FRAME COMPARISON

Box-force error decreased; impact and support errors remain

On 160 identical dynamic frames, box-force MSE decreased from 59.15 to 32.71 N²: **44.7% lower**.
The new parameters were selected using different calibration episodes.



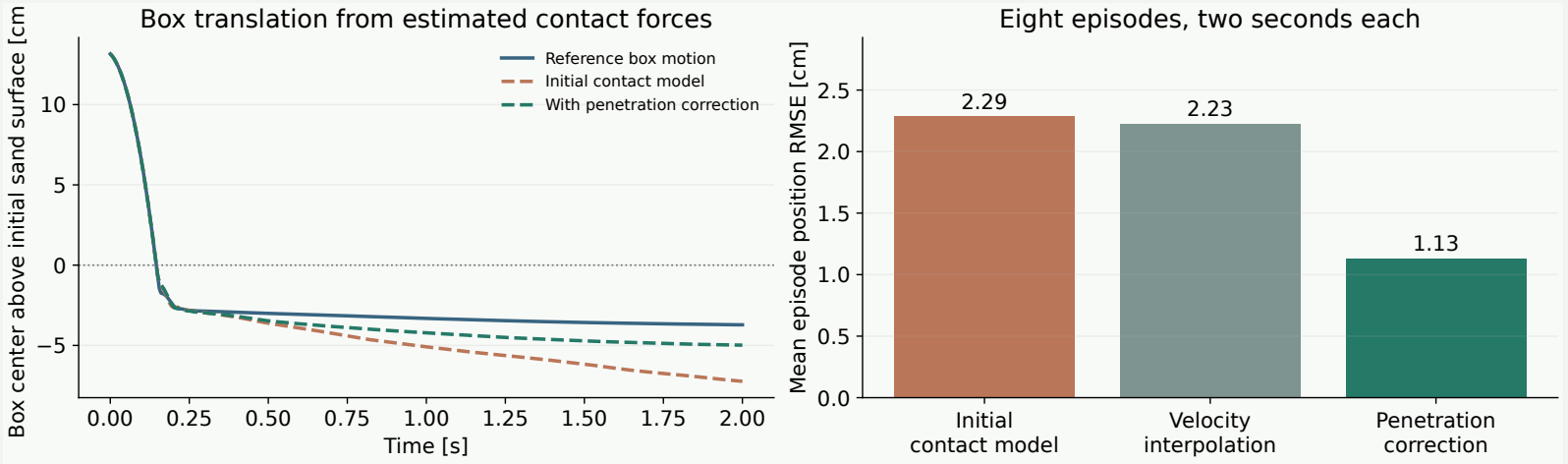
Separate dense test: 8 episodes, 3,200 transitions. Force-vector RMSE 10.69 N; relative force error 39.4%.
Steady upward force is approximately 12–15 N; the reference is 19.62 N.

Normal-contact fit: zero contact offset, effective contact mass = 10 × particle mass, start-of-step geometry. Median CPU time ≈ 1.1 ms for one box.
Coulomb-friction variant: test force RMSE 14.46 N, torque RMSE 0.320 N m; normal-only torque RMSE 0.398 N m.
Earlier inverse models estimate all six bodies; the new estimator handles the dynamic box only. Latest progress

OFFLINE REPLAY

Contact correction improves box translation

Start from the recorded initial box state. Estimate its contact force at each step, then integrate its motion.
Particle motion remains the recorded reference; no MPM solver runs during this replay.



Mean position RMSE: 2.29 → 1.13 cm. Mean orientation RMSE: 9.57 → 8.93°.
The revised contact model also lowers force-vector RMSE from 10.69 to 9.85 N on the same 3,200 steps.

Correction: estimate particle velocity at the box surface and add a separating velocity proportional to penetration depth.
Eight 2-second episodes. Reference-force replay gives 1.34 µm mean position RMSE, checking force timing and integration.
Recorded particles cannot respond to box-motion errors. A full coupled NeRD rollout remains untested.

NUMERICAL CHECK

Force balance accepts several box reactions

Hold every recorded particle position and acceleration fixed. Change the box load and solve again for internal particle forces and reactions from the rigid ground and walls.

```
upward support from ground and walls − upward force on box = 256.74 N
```

UPWARD PARTICLE FORCE ON BOX	UPWARD GROUND-AND-WALL FORCE ON PARTICLES	PARTICLE-BALANCE RESIDUAL
0.00 N	256.74 N	0.0067%
13.15 N	269.89 N	0.0074%
26.31 N	283.04 N	0.0082%

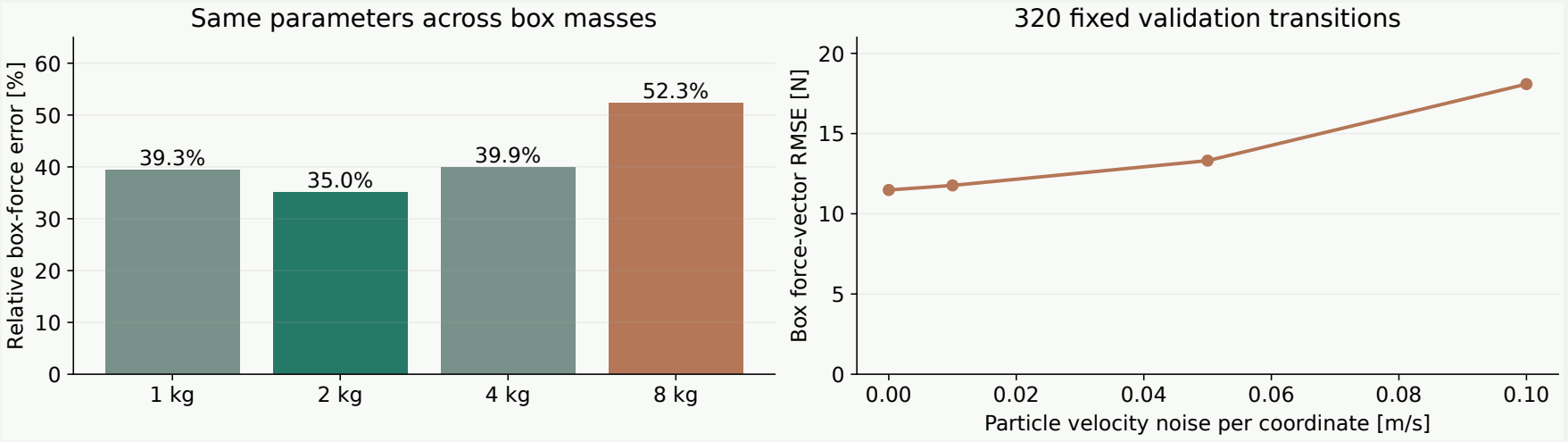
Particle balance alone accepts all three. Box dynamics and contact equations must select a physically consistent reaction. The recorded upward box reaction at this step is 19.95 N.

One recorded settling state; all 500 particle equations are solved. Internal forces are compressive and central.
Residual = norm of particle force imbalance / norm of individual particle weights. It does not measure box-force accuracy.

32 NEW EPISODES

Force accuracy remains sensitive to mass and particle errors

2 kg box: force-vector RMSE 9.41 N; component MSE 29.50 N²; torque-vector RMSE 0.344 N m.
12,800 transitions. The force RMSE's episode-bootstrap 95% interval is 8.91–9.87 N.



Remaining limitation: the approximation is calibrated on one material and one box geometry.
Particle velocity errors directly affect the recovered force; no learned-motion accuracy has been demonstrated.

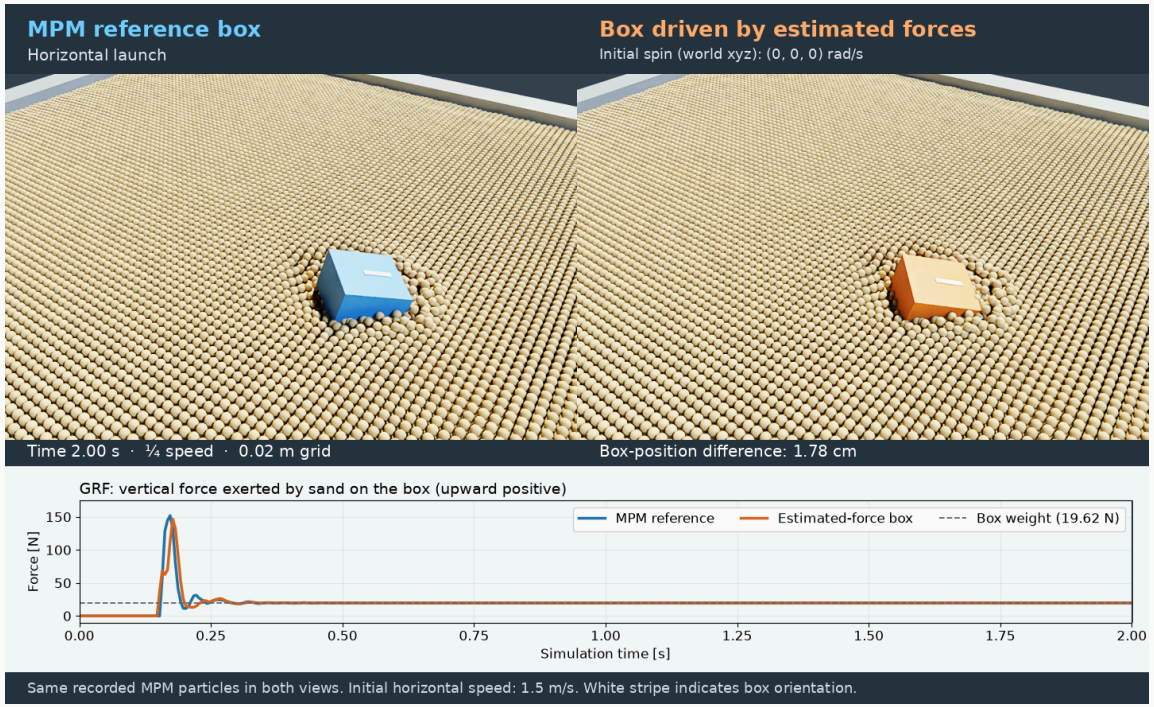
Frictional contact with affine surface velocity and penetration correction. Parameters remain fixed across all mass tests.
4 kg and 8 kg cases also include direct rigid contacts in 1.5% and 29.8% of transitions. Force targets are particle-on-box reactions only.
Noise study: independent Gaussian error per velocity coordinate; exact box state; a separate 320-step validation subset.

4 NEW MOTIONS

Horizontal and spinning box launches

Motion Horizontal launch

v_0 = (1.5, 0, −0.1) m/s; ω_0 = (0, 0, 0) rad/s (world xyz).



Position RMSE 1.36 cm · Orientation RMSE 5.32° · Force RMSE 7.91 N

Both views use recorded MPM particles; box contact forces, torques, and motion differ.

Four launches share the same initial box pose. The box is freely dynamic after reset; model parameters are unchanged.
2 × 2 × 0.3 m sand bed · 0.02 m grid · 296,875 particles · 25 MPM iterations. GRF is the upward sand-on-box force.
GRF, horizontal force, and torque profiles · Open selected video · All four results

MEASURED TRADEOFF

Lower mean error; impact peaks remain difficult

Retain the previous estimated contact impulse and update it at every step. Start from zero impulse, and clear the history when contact separates. No measured force initializes this state.

```
material velocity correction = (new impulse − α × previous impulse) / effective mass
```

CONTACT MODEL	FORCE RMSE [N]	PEAK-FORCE RMSE [N]	BOX POSITION RMSE [CM]
No impulse history	9.41	40.55	0.72
History + recalibrated parameters	6.66	62.22	0.38

Average accuracy improves, but peak accuracy worsens.
The larger overnight study evaluates both objectives instead of selecting a model from mean error alone.

Same 32 development episodes, 12,800 transitions. Force component MSE: 29.50 → 14.80 N².
Peak error is upward particle-on-box force error at the reference peak time. Position error is the mean over force-driven box replays.
Recorded particle motion is used throughout. This contact law is a calibrated approximation, not a granular constitutive model.

LIVE RESULTS

Mean force error and impact peaks are evaluated separately

Validation: 32 episodes, 400 transitions each. Parameters are selected using separate fitting episodes.
Peak error is measured at the time of the reference upward-force peak.

MODEL	FORCE RMSE [N]	PEAK RMSE [N]	POSITION RMSE [CM]	ORIENTATION RMSE [°]
No impulse history · lowest peak error	9.32	38.33	0.63	3.71
Initial history model	6.77	60.01	0.27	2.57
Candidate 1 · lowest mean error	6.73	58.35	0.26	2.50

> Force estimates use predicted material motion and contact-impulse history. The measurements here use recorded particles; they do not establish a coupled NeRD rollout.

Validation replay uses eight episodes; final replay uses eligible test episodes without direct rigid contact. Counts appear on the progress page.
Particles follow their recorded trajectories. No MPM solver or force network runs in reconstruction or replay.
The study is paused while the method is reviewed. Published values are aggregate measurements; recordings remain local.

IMPLEMENTED · EXPERIMENTAL

One step: rigid motion → particles → contact force

Input: particle positions and velocity history; body poses and velocities; previous contact wrench; material properties and rigid geometry.

1. Advance the rigid body

Apply gravity and the previous estimated contact force and torque.

$$V_{t+1} = V_t + \Delta t\,(g + F_{t-1} / m)$$

Output: updated body pose and linear/angular velocity.

2. Predict all particle positions

The GNN uses neighboring particles and each body's signed distance, normal, and relative surface velocity.

$$\hat{x}_{t+1} = x_t + \Delta t\,(v_t + \Delta t\, g) + \text{GNN correction}$$

3. Obtain particle velocities

$$\hat{v}_{t+1} = (\hat{x}_{t+1} - x_t) / \Delta t$$

Optional ablation: project predicted particles outside static rigid geometry, then recompute velocity.

4. Estimate the next body wrench

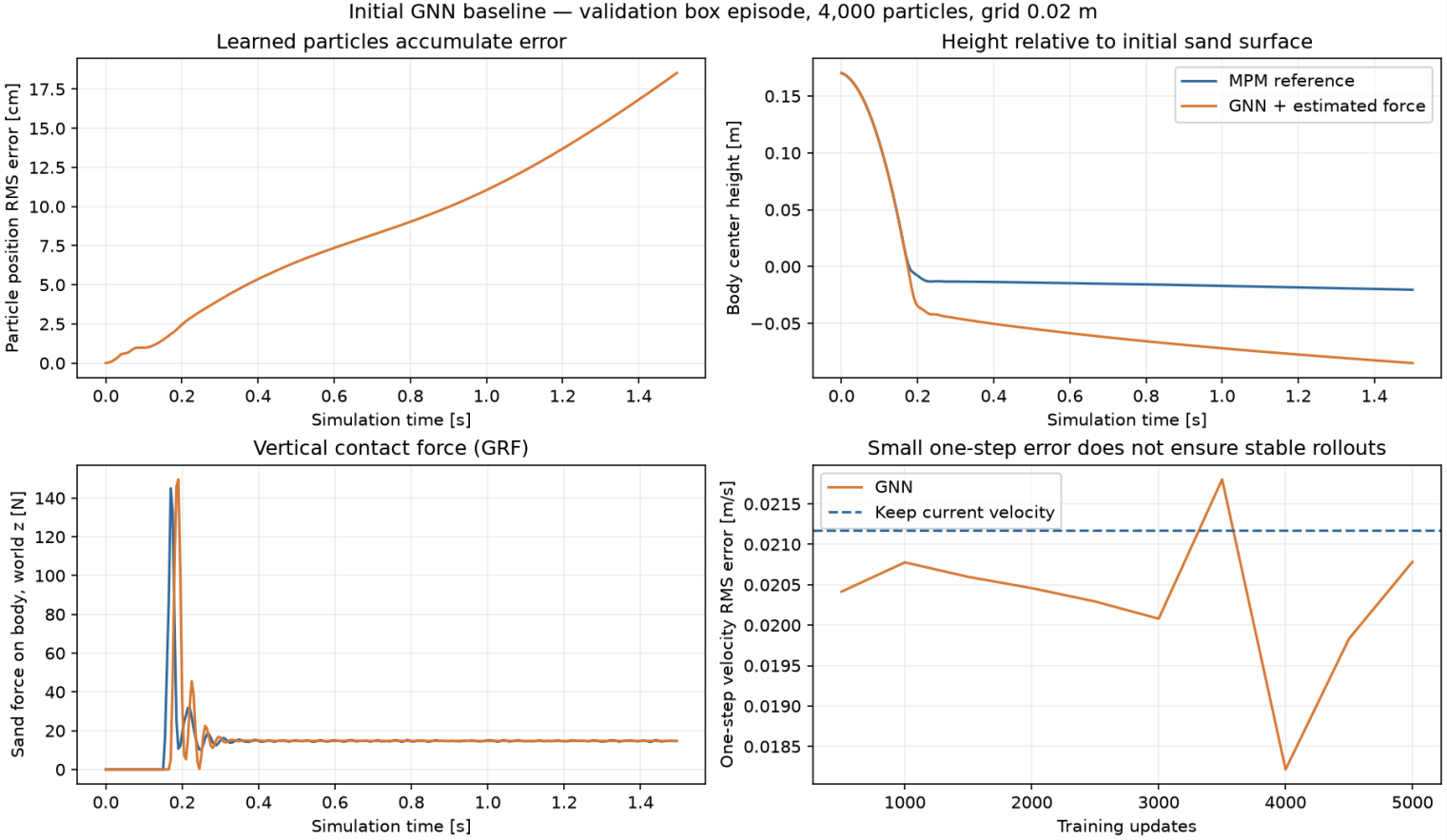
The analytical contact solver uses predicted particle velocities and the body's state at t.

$$F_t = \Sigma\, J_i / \Delta t \quad \cdot \quad \tau_t = \Sigma\,(r_i \times J_i) / \Delta t$$

Output: next particle/body states and a wrench for the following rigid step. No MPM solve or force network at inference. Live measurements →

INITIAL BASELINE FAILS

Small one-step errors can accumulate into large drift



What was tested

4,000 particles · 0.02 m MPM grid · 300 steps · 1.5 seconds.

48 training, 6 validation, and 12 reserved test episodes. Box, sphere, and cylinder geometries.

What failed

The initial network lets particles pass through the rigid ground beneath the sand.

Longer one-step training improves local accuracy but worsens drift. Rollout training and explicit static-contact correction are being evaluated.

The figure uses genuinely predicted particles and estimated forces. No reference particle states are injected after initialization. See updated comparisons and failures →

GEOMETRY ABLATION

One signed-distance interface supports different shapes

Query particle position → signed distance φ, outward normal n, and closest surface point.

$\varphi > 0$ outside the rigid body · $s \approx x - \varphi\, n$ · $n \approx \nabla\varphi / \|\nabla\varphi\|$

GEOMETRY	ANALYTIC FORCE RMSE	FORMULA → SDF	MESH → SDF	MESH-SDF DIFFERENCE
Box	5.19 N	5.16 N	5.16 N	0.18 N RMS
Sphere	6.35 N	6.44 N	6.56 N	0.44 N RMS
Cylinder	5.08 N	5.13 N	5.15 N	0.13 N RMS

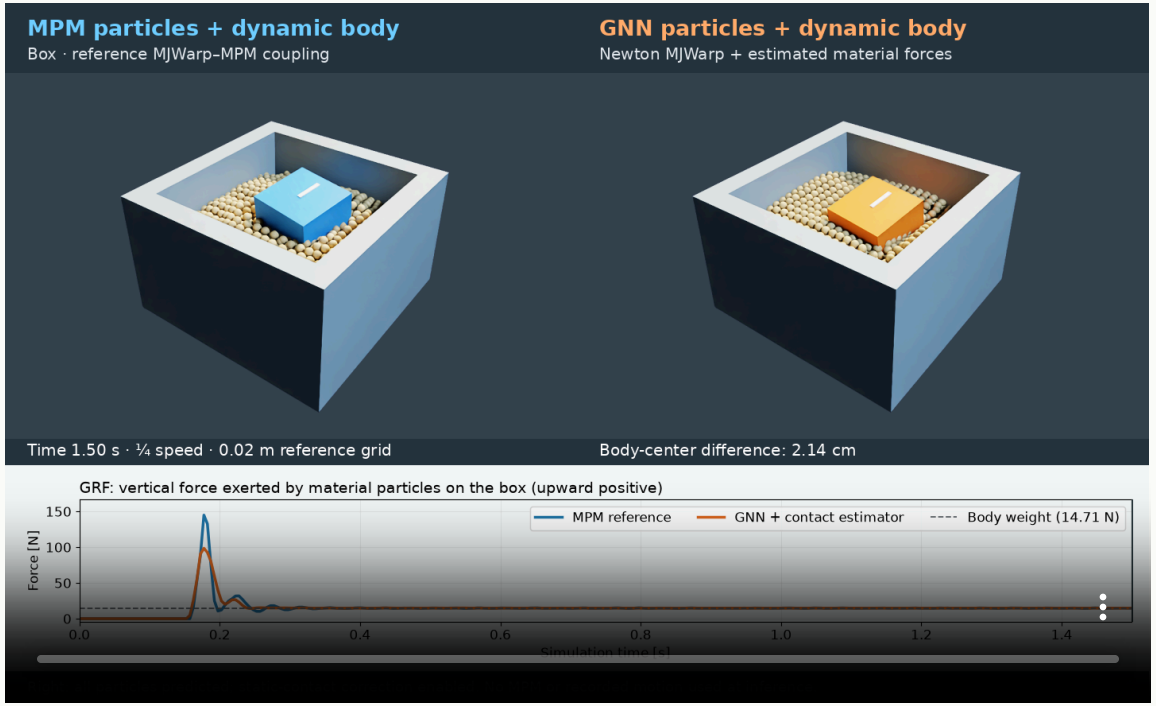
Recorded particle and rigid-body states isolate geometry error. Each shape uses 30 sampled validation transitions. Both sampled fields use 5 mm spacing, separate from the 0.02 m MPM grid. The last column compares mesh-SDF forces with analytic forces.

Closed-mesh SDF generation is implemented; a concave-prism check preserves its notch. Learned accuracy on unseen mesh shapes remains untested. Mesh illustration and details →

COUPLED VALIDATION

Both particle motion and body motion are predicted

Left: dynamic MJWarp–MPM reference. Right: GNN particles, analytical contact forces, and Newton rigid dynamics.



Six validation rollouts: particle error 1.42 cm · body-position error 1.76 cm · force error 5.24 N.
Impact remains underestimated in this box episode: peak upward force 98.5 N versus 145.0 N reference.

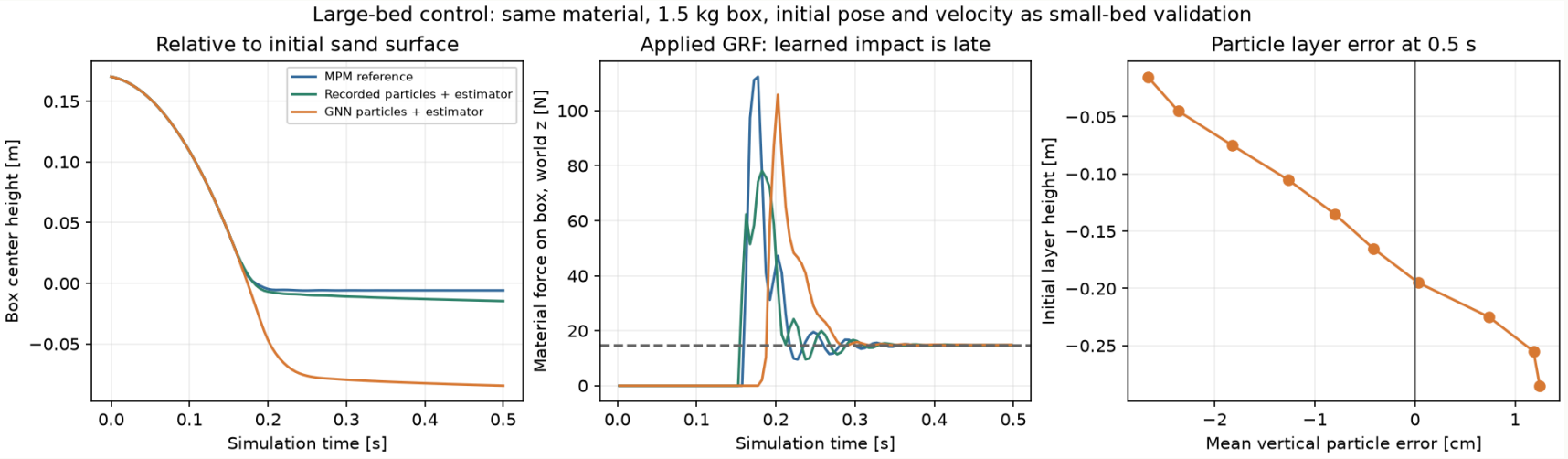
Four-step GNN training · 4,000 particles · 0.02 m reference grid · 1.5 seconds · no MPM at inference.
Particles are projected outside static rigid geometry. No reference particle or body motion is supplied after initialization.

Force and torque profiles · Box, cylinder, sphere, and current training results →

CONTROLLED DIAGNOSTIC

Particle-model error delays the body impact

Same material, 1.5 kg box, and initial body state. Larger sand bed: 2 × 2 × 0.3 m.



100-STEP COMPARISON (0.5 S)	BOX POSITION RMSE	FORCE RMSE
Small bed · GNN particles + estimator	0.64 cm	9.53 N
Large bed · recorded particles + estimator	0.49 cm	11.63 N
Large bed · GNN particles + estimator	5.72 cm	24.88 N

All 296,875 particles are updated; no region selector. Static-contact correction is enabled.

The initial top particle layer is 2.68 cm too low on average at 0.5 s. Accurate weight support does not establish accurate deformation.

RESEARCH STATUS

The hypothesis is clear; the particle reconstruction is open

Established by the sources

- Cable NeRD separates learned motion from analytical reactions.
- Newton exposes collider identity and body-reaction coupling.
- L-GBND represents terrain–tool interactions with typed nodes.

To establish experimentally

- A suitable material law and reconstruction discretization.
- Sufficient particle state for stable inference.
- Accurate reactions from imperfect learned motion.
- A net speed benefit for the coupled simulation.

[1] Cable NeRD · slides 23–24

[2] L-GBND · Liu, Li & Hauser

[3] L-GBND · released architecture

[4] XPBD · Macklin et al.

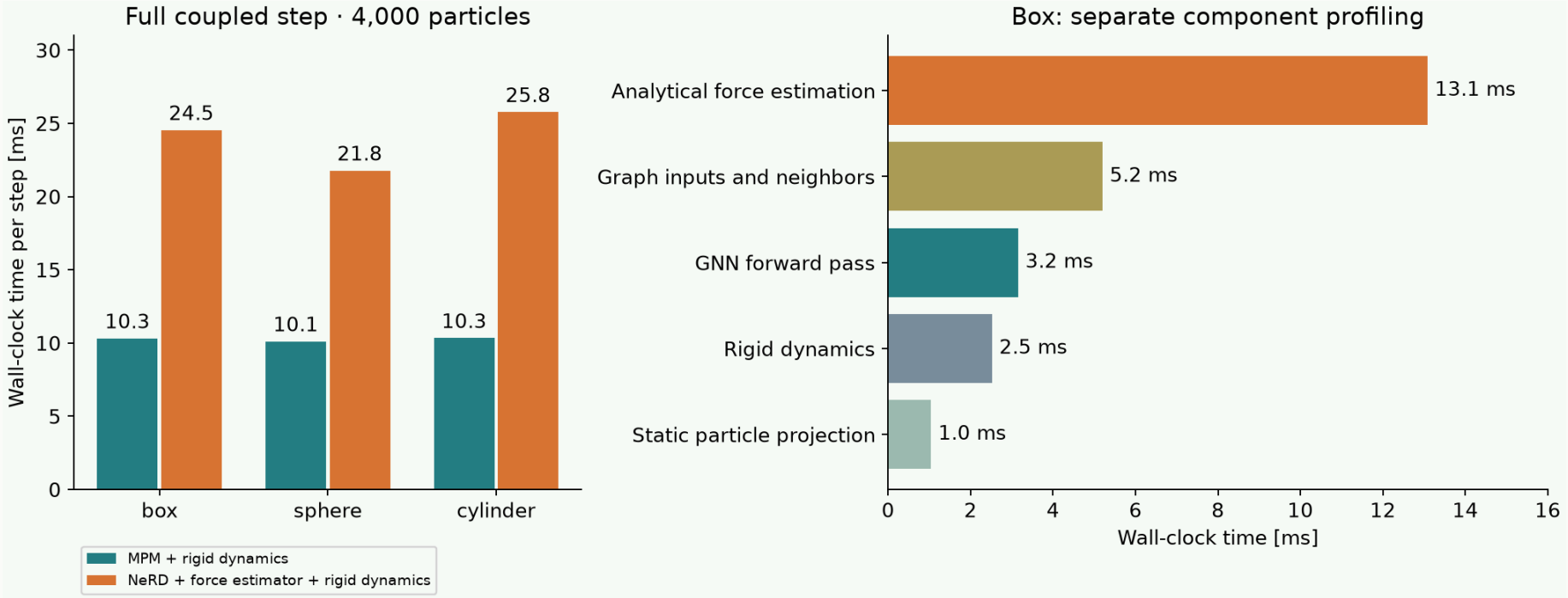
[5] Newton · coupled example

[6] Newton · implicit MPM solver

MEASURED

Earlier baseline: force estimation limited speed

4,000 particles · 5 ms timestep · RTX 5090 · no rendering or recording.



Matched initial states; one warmup trajectory and three 300-step measured trajectories. MPM: 25-iteration limit. Component profiling is measured separately.

Interactive play: right-drag the rigid object, Space to pause, Reset to restore the scene.

Commands, force plots, timing procedure, and limitations →

SELECTED CHECKPOINT

How the particle model computes one step

Implementation: particle_model.py, graph_inputs(), train_particles.py, and play.py. Selected model: 64 channels, 3 message-passing layers, 3 velocity frames, 4-step training.

G1 PLAY IS OOD

The selected model learned a small, fixed-resolution scene

CONDITION	SELECTED CHECKPOINT TRAINING	G1 PLAY EXPERIMENT
Sand bed	0.32 × 0.32 × 0.16 m	2 × 2 × 0.3 m
Particle count	4,000	39,690 / 296,875
MPM grid → particle spacing	0.02 m → 0.016 m	0.04 m → ~0.032 m; or 0.02 m → ~0.016 m
Box dimensions	12 × 12 × 8 cm	17 × 6 × 1 cm
Interaction	Passive launches; 1.5 s episodes	Mouse forces; arbitrary duration

48 training episodes: 4 fixed material/mass combinations × 3 shapes × 4 episodes. The later 156-episode collection was used by other models, not this selected checkpoint.

Brown particle color is only a visual change. Bed size, particle spacing, body dimensions, and applied forces change the model inputs and dynamics. Switching to 0.02 m does not remove the other distribution shifts.

Default play restored to recorded validation geometry. Use `--scene g1-testbed` explicitly for OOD tests.

Exact materials, launch conditions, checkpoint lineage, and missing coverage →

MEASURED

Physics stays on GPU; PyTorch runs the network

4,000 particles · RTX 5090 · 5 ms timestep · **15.73 ms per complete step**

STAGE	IMPLEMENTATION	MEAN [MS]
Rigid-body dynamics	Warp / MJWarp	0.23
Particle neighbors and graph edges	Warp	1.13
Model input features	Warp	0.02
GNN encoding, message passing, decoding	PyTorch	6.12
Particle integration, tray correction, history	Warp	0.06
Particle velocity-gradient fit	Warp	4.54
Contact setup and impulse solve	Warp	3.53

Zero device-to-host copies in the step. Rigid substep captured; full-step capture remains pending.
20 physics/parity tests pass. Same checkpoint and force equations; no retraining.

Whole-step timing: 900 synchronized samples. Stage CUDA events: separate 300-step rollout.

All ten stages, transfer audit, numerical agreement, and reproduction commands →